

OBJECTIVE

Results-driven professional in AI and Data Science with a focus on developing intelligent solutions to address practical challenges. Currently pursuing a Bachelor of Technology in Artificial Intelligence at Saranathan College of Engineering, with an expected completion in 2027. Committed to applying technical knowledge to create impactful applications in the field. Skills include data analysis, machine learning, and problem-solving.

EDUCATION

Saranathan College of Engineering
Bachelor of Technology; GPA: 8.33
Artificial Intelligence & Data Science

Trichy, India

Sep 2023 - Dec 2027

SKILLS SUMMARY

- **Languages:** C, Python, SQL, java
- **Frameworks:** Pandas, Numpy, Scikit-Learn, Pytorch,Tensorflow, Langchain, Langgraph, RAG
- **Tools:** Power BI, Excel, MySQL
- **Soft Skills:** Rapport Building, Leadership, Time Management, Excellent communication.

PROJECTS

Project: Air Pollution Forecasting System using AI/ML

- Developed an automated AI-driven forecasting system to predict hazardous gas concentrations, specifically Nitrogen Dioxide (\$NO_2\$) and ground-level Ozone (\$O_3\$), in rapidly growing urban centers.
- Technologies used: Python, FastAPI, Scikit-learn, Pandas, Uvicorn.
- Implemented a system that:
 - Models non-linear relationships between complex meteorological variables and satellite-derived atmospheric data.
 - Delivers 24-48 hour air quality forecasts at hourly intervals through a high-performance REST API.
 - Integrates real-time data streams to ensure the model stays current with changing weather patterns.

Project: Speech Emotion Recognition (SER) System

- Developed a deep learning system to classify human emotions from audio into 8 categories using recurrent neural networks.
- Technologies used: Python, TensorFlow, Keras, Librosa, Scikit-learn, NumPy.
- Implemented a system that:
 - Feature Engineering: Extracts 40-dimensional MFCCs from raw audio to capture essential acoustic characteristics.
 - LSTM Architecture: Utilizes Long Short-Term Memory layers to model temporal dependencies in speech sequences.
 - Model Optimization: Employs Batch Normalization and Dropout to improve generalization and training stability.
 - Inference Pipeline: Processes unseen audio files for real-time classification using a softmax output layer.

CERTIFICATES

Machine Learning Specialization (Deeplearnig.Ai) |[CERTIFICATE](#)

- Learned about the supervised learning, unsupervised learning and neural networks.
- Hands on experience with Python,Numpy,Scikit-learn, Tensorflow and Keras.
Model evaluation and tuning

Infosys Springboard – Introduction to Artificial Intelligence | [CERTIFICATE](#)

- Completed a foundational course covering Artificial Intelligence concepts, intelligent systems, and AI applications.
- Learned basics of machine learning, problem-solving techniques, and data-driven decision making.

Infosys Springboard – Introduction to Data Science | [CERTIFICATE](#)

- Gained knowledge in data analysis, data preprocessing, statistical methods, and data visualization.
- Developed understanding of extracting insights from structured datasets using analytical techniques.

Infosys Springboard – Introduction to Deep Learning | [CERTIFICATE](#)

- Learned deep learning fundamentals including neural networks, activation functions, and model training concepts.
- Explored practical applications of deep learning in AI systems.

Infosys Springboard – Python for Data Science | [CERTIFICATE](#)

- Completed training in Python programming for data analysis and scientific computing.
- Worked with Python libraries for data manipulation, visualization, and analytics workflows.